



TIANHAO HUANG

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EDUCATION

Shanghai Jiao Tong University

Sep. 2025 – Present

Ph.D. student in Computer Science.

Advisor: Prof. Chao Li.

Research: Computer architecture and performance modeling.

Shanghai Jiao Tong University

Sep. 2021 – 2025

Bachelor's degree in Information Security. GPA: 3.9/4.3; average score: 90.42/100.

PROJECT EXPERIENCE

ARM Instruction Microbenchmark Suite

Nov. 2025 – Present

Shanghai Jiao Tong University and Huawei

- Built ARM instruction microbenchmarks to measure instruction throughput and latency.
- Implemented execution-port mapping inference with adaptive instruction-pair tests and interference-resilient constraints.

Cross-CPU Performance Modeling for Production Workloads

Apr. 2025 – Aug. 2026

Shanghai Jiao Tong University and Alibaba Cloud

- Decoupled workload and CPU representations using hardware traces and simulator configurations.
- Combined simulation and limited measurements to improve cross-CPU prediction accuracy.

GPU–NPU Co-design with Shared SRAM

Sep. 2024 – Aug. 2025

Shanghai Jiao Tong University

- Designed shared SRAM configurable as GPU last-level cache or NPU scratchpad for on-chip data exchange.
- Implemented flexible SRAM and bandwidth allocation and overlapped GPU–NPU operator execution.

BEV Perception on Dual SoCs

Sep. 2023 – Aug. 2024

Shanghai Jiao Tong University

- Deployed BEV-centric perception networks on a dual-SoC platform using RDMA communication.
- Split networks into multiple TensorRT engines (plans) to support deployment across SoCs.

End-to-End Multimodal Benchmarking

Jan. 2023 – Jun. 2024

Shanghai Jiao Tong University

- Studied multimodal DNN architectures and analyzed their execution patterns on CPUs and GPUs.
- Customized multimodal neural network inference workloads of different sizes for system and architecture evaluation.

PUBLICATIONS

1. Xinkai Wang, Chao Li, **Tianhao Huang**, Jinyang Guo, Xiaofeng Hou, Guoyao Xu, Yuemin Wu, Minyi Guo. *Intra-service tracing for shared and stressed datacenters: efficient management and intelligent analysis*. *Scientia Sinica Informationis*, 56(7): 1620–1634, 2026 (in Chinese).
2. **Tianhao Huang**, Lingyu Sun, Chao Li, Xiaofeng Hou, Yaqian Zhao, Jingwen Leng, Li Li, Minyi Guo. *Repurpose Accel-Sim for Next Generation NVIDIA Jetson GPU Architectural Design*. *IEEE/ACM ISLPED*, 2025, pp. 1–7.
3. Lingyu Sun, Chao Li, Xiaofeng Hou, **Tianhao Huang**, Cheng Xu, Xinkai Wang, Guangjun Bao, Bingchuan Sun, Shibo Rui, Minyi Guo. *Jigsaw: Taming BEV-centric Perception on Dual-SoC for Autonomous Driving*. *IEEE RTSS*, 2024, pp. 280–293.
4. **Tianhao Huang**, Lingyu Sun, Xiaofeng Hou, Xiaozhi Zhu, Xinfeng Xia, Yutong Wang, Mingxi Chen, Chao Li. *Sub-model Parallelism: A Scale-out Deployment Method for Large Multi-modal DNNs*. *IEEE CCGrid*, 2024, pp. 651–657.
5. **Tianhao Huang**, Xiaozhi Zhu, Mo Niu. *An End-to-End Benchmarking Tool for Analyzing the Hardware-Software Implications of Multi-modal DNNs*. *ACM SIGMETRICS Performance Evaluation Review*, 51(3): 25–27, 2024.
6. Cheng Xu, Xiaofeng Hou, Jiacheng Liu, Chao Li, **Tianhao Huang**, Xiaozhi Zhu, Mo Niu, Lingyu Sun, Peng Tang, Tongqiao Xu, Kwang-Ting Cheng, Minyi Guo. *MMBench: Benchmarking End-to-End Multi-modal DNNs and Understanding Their Hardware-Software Implications*. *IEEE IISWC*, 2023, pp. 154–166.

TALKS

- *Repurpose Accel-Sim for Next Generation NVIDIA Jetson GPU Architectural Design*. *IEEE/ACM ISLPED*, August 4–6, 2025 (in person).
- *Sub-model Parallelism: A Scale-out Deployment Method for Large Multi-modal DNNs*. *IEEE/ACM CCGrid*, Philadelphia, USA, May 6–9, 2024 (in person).
- *An End-to-End Benchmarking Tool for Analyzing the Hardware-Software Implications of Multi-modal DNNs*. *ACM SIGMETRICS*, Orlando, Florida, USA, June 19–22, 2023 (online).

OPEN-SOURCE SOFTWARE

- **MMBench**: An end-to-end benchmarking tool for analyzing the system and architecture implications of multimodal DNNs.
<https://github.com/xfhelen/MMBench>
- **Accel-Sim-J**: Repurposing Accel-Sim for next-generation NVIDIA Jetson GPU architectural design.
<https://github.com/sjtu-SAIL-lab/accel-sim-framework/tree/Accel-sim-J>

HONORS AND AWARDS

- Shanghai Municipal Scholarship (2024)
- Second Place, TCSC SCALE Challenge, *IEEE/ACM CCGrid* (2024)
- First Place, Undergraduate Category, *ACM SIGMETRICS Student Research Competition* (2023)
- SJTU Undergraduate Scholarships (academic years 2021–2022 and 2022–2023)
- Meritorious Winner, Mathematical Contest in Modeling (MCM, 2023)
- Elite Award, AI University Group, SJTU–Winning Health Smart Medical Challenge (2023)

SKILLS AND LANGUAGES

Programming and tools: C++, Python, Microsoft Office; CCF CSP: 355.

Languages: Mandarin Chinese (native); English (CET-6: 544).